Workflow Automation for RNA-seq Using Snakemake

Author: Katarzyna Zielińska

Year: 2026

Introduction

RNA-seq analysis consists of multiple computational steps that need to be performed in a defined and reproducible order. Manual execution of these steps can make workflows difficult to reproduce and maintain. This project focuses on automating an RNA-seq alignment workflow using **Snakemake**, integrating **HISAT2** for genome alignment and **SAMtools** for BAM processing.

Project Objective

The main objective was to develop a reproducible workflow that automatically:

- processes paired-end trimmed RNA-seq reads
- aligns reads to the GRCh38 reference genome
- sorts the resulting alignments
- indexes the final BAM file
- manages dependencies between workflow steps using Snakemake

The project also demonstrates the use of configuration files and workflow validation.

Dataset

The workflow uses paired-end RNA-seq data:

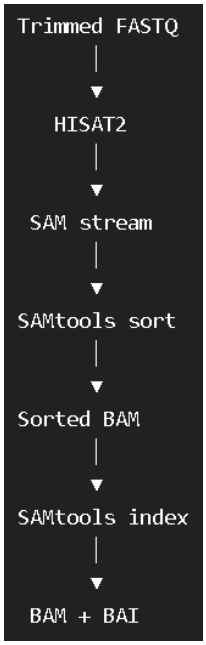
Information	Value
Organism	*Homo sapiens*
Reference genome	GRCh38
Data type	Paired-end RNA-seq
Input	Trimmed FASTQ
Alignment	HISAT2
BAM processing	SAMtools

Input files:

```
data/trimmed/
├── sample_R1_trimmed.fastq
└── sample_R2_trimmed.fastq
```

A pre-built GRCh38 HISAT2 index was used for alignment and validated with hisat2-inspect-s.

Workflow



The workflow was implemented in a **Snakefile**, with computational parameters stored in config.yaml. The HISAT2 output is streamed directly into samtools sort, avoiding the creation of an intermediate SAM file.

Tools and Technologies

Workflow & Alignment

- Snakemake
- HISAT2
- SAMtools

Quality Control & Preprocessing

- FastQC
- fastp

These tools were used in the upstream RNA-seq preprocessing stages incorporated into the project workflow.

Environment

- Linux / Bash
- WSL2
- Graphviz

Implementation

The main Snakemake rule connects HISAT2 alignment with SAMtools sorting:

```
hisat2 \
    -p {threads} \
    -x {params.index} \
    -1 {input.r1} \
    -2 {input.r2} \
    2> {output.log} \
| samtools sort \
    -@ {threads} \
    -o {output.bam} \
```

A separate Snakemake rule performs BAM indexing:

```
samtools index {input.bam}
```

The number of threads is defined in config.yaml:

```
threads: 4
```

This allows computational resources to be changed without modifying the workflow itself.

Results

The workflow successfully generated a coordinate-sorted and indexed BAM file.
Alignment statistics

Metric	Result
Paired reads	28,499,041
Overall HISAT2 alignment rate	98.34%
Mapped reads	98.60%
Properly paired	96.27%
Singletons	1.34%
Final BAM size	~1.9 GB

Discussion

The workflow successfully automated the main RNA-seq alignment steps and established explicit dependencies between alignment, sorting and indexing. The high alignment rate of **98.34%** indicates that the majority of reads were successfully aligned to the GRCh38 reference genome. Using Snakemake makes the workflow reproducible and allows missing or incomplete outputs to be automatically regenerated. The direct HISAT2-to-SAMtools pipeline also avoids storing a large intermediate SAM file.

Conclusion

This project demonstrates the implementation of a reproducible RNA-seq alignment workflow using Snakemake, HISAT2 and SAMtools. The workflow successfully transforms paired-end trimmed FASTQ files into a coordinate-sorted and indexed BAM file while automatically managing dependencies between computational steps.

The project demonstrates practical skills in:

- RNA-seq workflow development,
- genome alignment,
- BAM/SAM processing,
- workflow automation,
- reproducible computational analysis,
- Linux/Bash command-line workflows.

Repository Structure

```
05_Workflow-Automation-Snakemake/
│
├── Snakefile
├── config.yaml
│
├── data/
│   └── trimmed/
│
├── docs/
│
├── figures/
│   └── workflow_dag.png
│
├── genome/
│   └── hisat2_index/
│
└── results/
    ├── fastp/
    ├── qc/
    └── hisat2/
        ├── sample_Aligned.sortedByCoord.out.bam
        ├── sample_Aligned.sortedByCoord.out.bam.bai
        └── sample_Log.final.out
```

References

1. Polański A., Podstawy bioinformatyki, Wydawnictwo Polsko-Japońskiej Wyższej Szkoły Technik Komputerowych, Warszawa, 2010.
2. Lesk A. M., Wprowadzenie do bioinformatyki, Wydawnictwo Naukowe PWN, Warszawa, 2019.
3. Wang X., Next-Generation Sequencing Data Analysis, 2nd ed., CRC Press, 2024.
4. Korpelainen E., Tuimala J., Somervuo P., Huss M., Wong G., RNA-seq Data Analysis: A Practical Approach, CRC Press, 2015.
5. Azad R. K. (ed.), Transcriptome Data Analysis, Methods in Molecular Biology, vol. 2812, Springer, 2024.
6. A Guide to Applied Machine Learning for Biologists, Springer, 2023.
7. Géron A., Hands-On Machine Learning with Scikit-Learn and PyTorch: Concepts, Tools, and Techniques to Build Intelligent Systems, O'Reilly Media, 2025.
8. Alkhateeb A., Rueda L. (eds.), Machine Learning Methods for Multi-Omics Data Integration, Springer, 2023.
9. Ravarani C., Latysheva N., Deep Learning for Biology, O'Reilly Media, 2025.
10. Chollet F., Watson M., Deep Learning with Python, 3rd ed., Manning Publications, 2025.
11. Chollet F., Kalinowski T., Deep Learning with R, 3rd ed., Manning Publications, 2026.